

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem
Bounded source-aligned checkpoint — 19 July 2026

Authority. Corrected French lines 2796–2805; original printed p. 93; physical source-PDF p. 82; recomposed running p. 74. No printed-page marker occurs in this unit: marker 93 precedes it at line 2782 and marker 94 follows it at line 2813. Line 2806 is blank; line 2807 begins the next step and is the exact substantive continuation cursor. The compiled French reader is generated from the same corrected edition and is page/layout evidence, not independent corroboration.

The second step is to deduce (b) from (d).

Set

$$D(Z_q) = \{\dim \mathcal{O}_{X,y} \mid y \in Z_q\}.$$

By (d), for every $q \in \mathbb{Z}$ one has $q + i \notin D(Z_q)$. Applying Proposition VII.2.3, we see that

$$\mathcal{E}xt_Y^{q+i}(\mathcal{E}xt^q(\mathcal{F}, \mathcal{O}_X), \mathcal{O}_X) \quad \text{is coherent.}$$

The initial term of the spectral sequence of the preceding subsection is

$$E_2^{p,q} = \mathcal{E}xt_Y^p(\mathcal{E}xt^{-q}(\mathcal{F}, \mathcal{O}_X), \mathcal{O}_X).$$

It follows that $E_2^{p,q}$ is coherent for every $p, q \in \mathbb{Z}$ such that $p + q = i$. Now there are only finitely many pairs (p, q) such that $p + q = i$,^[S] and this spectral sequence converges to $\mathcal{H}_Y^*(\mathcal{F})$; whence the conclusion.

^[S]*Source note.* Taken literally for arbitrary integers, this sentence is false. In context it must be read as concerning the terms that can contribute, in view of the finite-projective-dimension setup above. The French authority is unchanged.